



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES, MIHINTALE
B.Sc. (General) Degree

Tutorial 2 – ALGEBRA – MAP 2302

1. Consider the following relations on the set of integers.

- I. $R_1 = \{(a, b) | a \leq b\}$
- II. $R_2 = \{(a, b) | a > b\}$
- III. $R_3 = \{(a, b) | a = b \text{ or } a = -b\}$
- IV. $R_4 = \{(a, b) | a = b\}$
- V. $R_5 = \{(a, b) | a = b + 1\}$
- VI. $R_6 = \{(a, b) | a + b \leq 3\}$

Which of these relations are reflexive, symmetric, anti-symmetric, or transitive?

2. Consider the following relations on $\{1, 2, 3, 4\}$:

- I. $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$
- II. $R_2 = \{(1,1), (1,2), (2,1)\}$
- III. $R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (3,4), (4,4)\}$
- IV. $R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$
- V. $R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$
- VI. $R_6 = \{(3,4)\}$

3. Let R_1 and R_2 be equivalence relations on a set S . show that the relation on R on S , defined by xRy iff xR_1y and xR_2y , is an equivalence relation. Is the relation $\sim$ on S , defined by $x \sim y$ iff xR_1y or xR_2y , an equivalence relation?
4. a. A relation R on $S = \{(x, y) | x, y \in \mathbb{Z}\}$ is defined by $(a, b)R(c, d)$ if $a^2 + d^2 = c^2 + b^2$. Is R an equivalence relation? Justify your answer.
b. show that relation R defined on the set of real numbers $\mathcal{R}$ by xRy if $x^2y - y^2x - 2(x - y) = 0$ is an equivalence relation on $\mathcal{R}$.
5. A relation on R on a set S satisfies the following two properties:
- I. R is reflexive.
 - II. for all $a, b, c \in S$, aRb , and bRc imply cRa

Prove that R is an equivalence relation.

6. a. Let R be the relation from $A = \{2,3,4,5,6\}$ to $B = \{3,5,8,12\}$ defined by “ a divides b ” for $a \in A$ and $b \in B$.
 b. Write R as a set of ordered pairs.
 Find the subset $\{a | (a, 12) \in R\}$ of A .
 Find R^{-1} .

b. let $X = \{1,2,3,\dots,8,9\}$. If $R = \{(x,y) | x,y \in X \text{ such that } x - y \text{ is a non-zero multiple of } 2\}$ and $S = \{(x,y) | x,y \in X \text{ such that } x - y \text{ is a non-zero multiple of } 3\}$. Find $R \cap S$ and $R \cup S$.

7.

- a. Let $A = \{1,2,3,4\}$. Consider the following relations on A .

$$R_1 = \{(1,2), (4,3), (2,2), (2,1), (3,1)\}$$

$$R_2 = \{(2,2), (3,2), (2,3), (3,3)\}$$

$$R_3 = \{(1,2)\}$$

Determine whether each relation is (i) symmetric (ii) transitive.

- b. Let $X = \{a, b, c, d, e, f, g\}$ and let

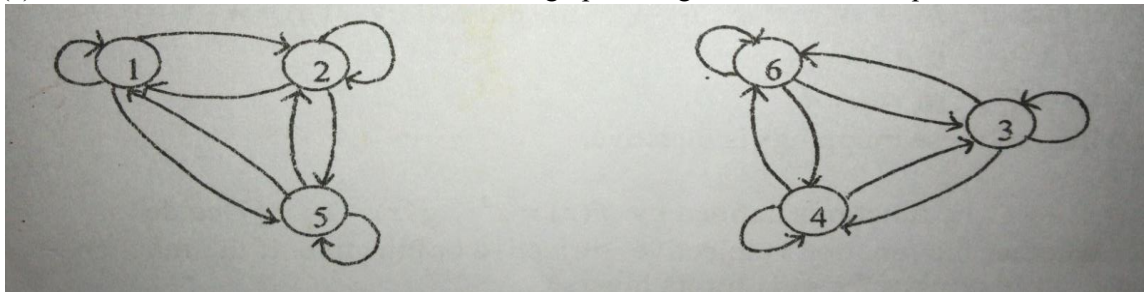
$$A_1 = \{a, b, c\} \quad A_2 = \{a, d, e, f\} \quad A_3 = \{g\}$$

$$B_1 = \{a, b, e\} \quad B_2 = \{c, f, g\} \quad B_3 = \{d\}$$

$$C_1 = \{e, f, g\} \quad A_2 = \{a, c, d\}$$

Which of the $\{A_1, A_2, A_3\}$, $\{B_1, B_2, B_3\}$, $\{C_1, C_2\}$ are partitions of X ?

8. (a) Determine whether the relation R whose digraph is as given below is an equivalence relation.



- (b) Determine whether the relation, whose matrix is $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$, is an equivalence relation.

9. (a) Which of the following subsets f of $A \times B$ are functions $f: A \rightarrow B$?

I. For $A = B = \mathbb{Z}$, $f(x) = |x|$.

II. For $A = B = \mathbb{R}$, $f(x^2) = x^3$

III. For $A = B = \{1,2,3,4\}$, $f = \{(2,3), (1,4)(2,1)(3,2)(4,4)\}$.

IV. For $A = \mathbb{Z}$, $B = \{0,1\}$, $f = \begin{cases} 1 & \text{if } x \text{ is odd} \\ 0 & \text{if } x \text{ is even} \end{cases}$

10. Determine the image of each of the following functions $f: A \rightarrow B$.

I. For $A = B = \{1,2,3,4\}$, $f = \{(2,3), (1,4)(3,2)(4,4)\}$

II. For $A = B = \{1,2,3,4,5,6,7,8\}$, $f = \text{largest of } x \text{ and } 4, f(4) = 4$

III. For $A = B = \mathbb{R}$, $f(x) = \frac{3x}{x^2+1}$.

IV. For $A = B = \mathbb{R}$, $f(x) = \begin{cases} 1 & \text{if } x \geq 2^x \\ 0 & \text{if } x < 2^x \end{cases}$

11. (a) Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be mappings, where X, Y, Z are arbitrary sets. Prove the following results about the composition mapping $g \circ f: X \rightarrow Z$:
- If f and g are injective the $g \circ f$ is injective.
 - If $g \circ f$ is injective then f is injective.
12. The two mappings $f: N \rightarrow N$ and $g: N \rightarrow N$ are defined by $f(n) = n + (-1)^n$ and $g(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ (n-1)/2 & \text{if } n \text{ is odd} \end{cases}$. Determine which of these mappings is bijective.
13. (a) Let $f, g: C \rightarrow C$ be functions defined by $f(z) = z^2, g(z) = 1 + z$. Decide in each case whether the function is injective, surjective or bijective. If the function is bijective, write down a formula for its inverse.
- (b) Let $N = Z^+ \cup \{0\}$. If $n \in Z^+$ show that $n = 2^r(2m+1)$ for unique $r, m \in N$. A function $f: Z^+ \rightarrow N \times N$ is defined by $f(n) = (r, m)$. Show that f is bijective.
14. (a) A function f is defined by $f(x) = (10 - k^2)x^2 - (k - 4)x + k$, where k is a given integer. For what values of k is f injective?
- (b) let n be a given integer, and let $f: Z \rightarrow Z$ be the function defined by $f(x) = x + n - xn$. Find the values of n for which f is (i) injective (ii) surjective. Also find f^{-1} when f is bijective.
15. Define reflexive symmetric, transitive, and anti-symmetric properties of a relation. Consider the following relations on the set of integers.
- $R_1 = \{(a, b) | a \neq b\}$
 - $R_2 = \{(a, b) | ab \leq 1\}$
 - $R_3 = \{(a, b) | a = b + 1 \text{ or } b = a + 1\}$

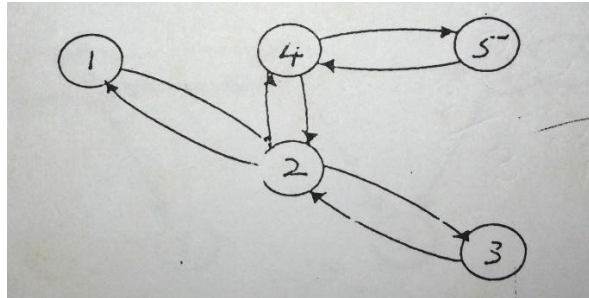
Decide in each case which of the above relations are reflexive, symmetric, transitive, and anti-symmetric. Justify your answers.

The functions $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ are defined such that $f(x) = x^2 + 1, g(x) = x^3 - x$, and $h(x) = 2^x$. Determine which of the above functions are one to one, onto correspondences. Justify your answers.

16. (a) let R be a relation on $A = \{a_1, a_2, a_3, \dots, a_n\}$. Prove that

- $M_{R^2} = M_R \odot M_R$
- $M_{R^2} = M_R$, then R is transitive.

(b) Let $A = \{1, 2, 3, 4, 5\}$ and let R be a relation on A whose digraph is shown below. Find R, M_{R^2} and M_{R^4} . Determine whether the relation R^2 is an equivalence relation on A .



Draw the digraph of the relation R^∞ on A . (Hint: draw 1, 2, 5, 4 in a horizontal line and draw 3 above 2).

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